

UNIT-3

Transport Layer:

- Second layer in TCP/IP and Fourth in OSI
- End to end layer used to deliver messages to a host

- Provides point to point connection, rather than hop to hop.

- Unit of data: segment

- 2 communication categories:

(i) Connection oriented

(ii) Connection less

- 2 protocols:
(i) TCP
(ii) UDP

- Responsibilities:
(i) Process the process delivery
(ii) End to end connection
(iii) Multiplexing & demultiplexing
(iv) Congestion control $\rightarrow$ prevent overflowing
(v) Data integrity & error correction
(vi) Flow control

Point to Point communication:

- It maps data between specific apps on each host using IP plus ports
- Transport layer delivers data from one application to another, not just host to host
- Ports: Each process has a port; segments carry source and destination ports to identify apps
- Socket: IP address + port = socket; defines exact endpoint on both sides
- Multiplex/Demultiplex: Many app streams share the network; receiver uses the destination port to hand data to right process

RIP = Routing Information Protocol

UDP-

Fixed = 8 bytes = 64 bits

UDP Header UDP Data

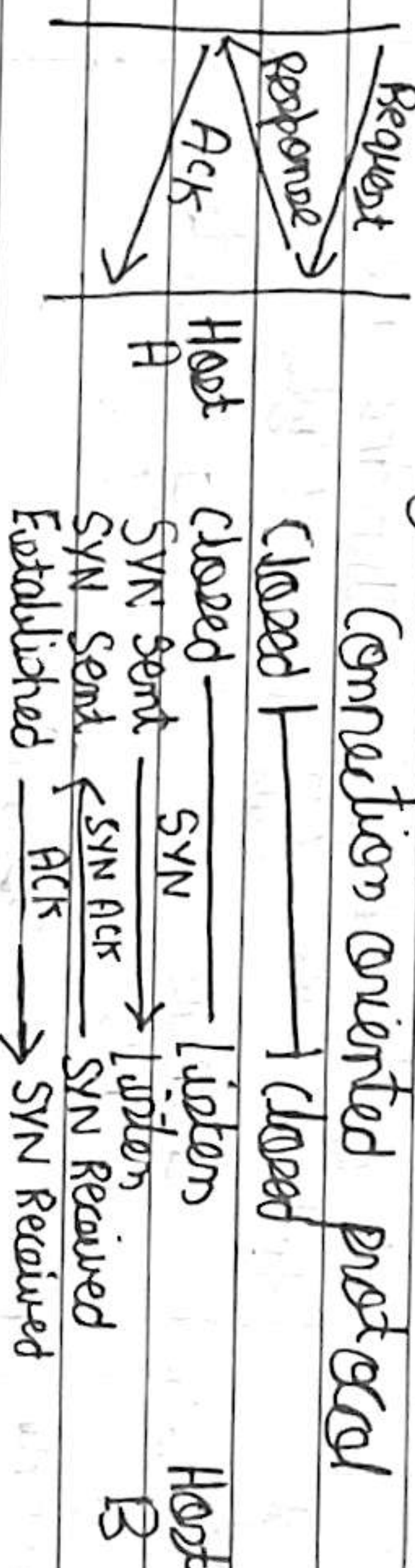
Source Port 16 bits	Destination Port 16 bits
Length 16 bits	Checksum 16 bits

- UDP header is 8 bytes long, followed by the data payload
- Application:

(i) DNS uses UDP for fast query/response lookups
(ii) DHCP (Dynamic Host Configuration Protocol) assigns IP addresses to devices
(iii) RIP
(iv) VoIP (Voice Over Internet Protocol)

TCP-

- 3 way handshaking method



- Lies between Internet and Application layer
- Prevents data overflow
- Main protocol for TCP/IP
- Uses checksum to detect complicated problems
- Features:
- (i) Segment Numbering
- (ii) Flow control: Sliding Window
- Stream oriented, reliable & connection oriented
- Data is divided into segments
- Sequence numbers help rearrange data in order
- End-to-end & end-to-end numbering of host segments

- 3 way Handshake :

- (i) SYN (client → server) with initial sequence number
- (ii) SYN + ACK (server → client) with its (ISN)
- (iii) ACK (client → server) own ISN

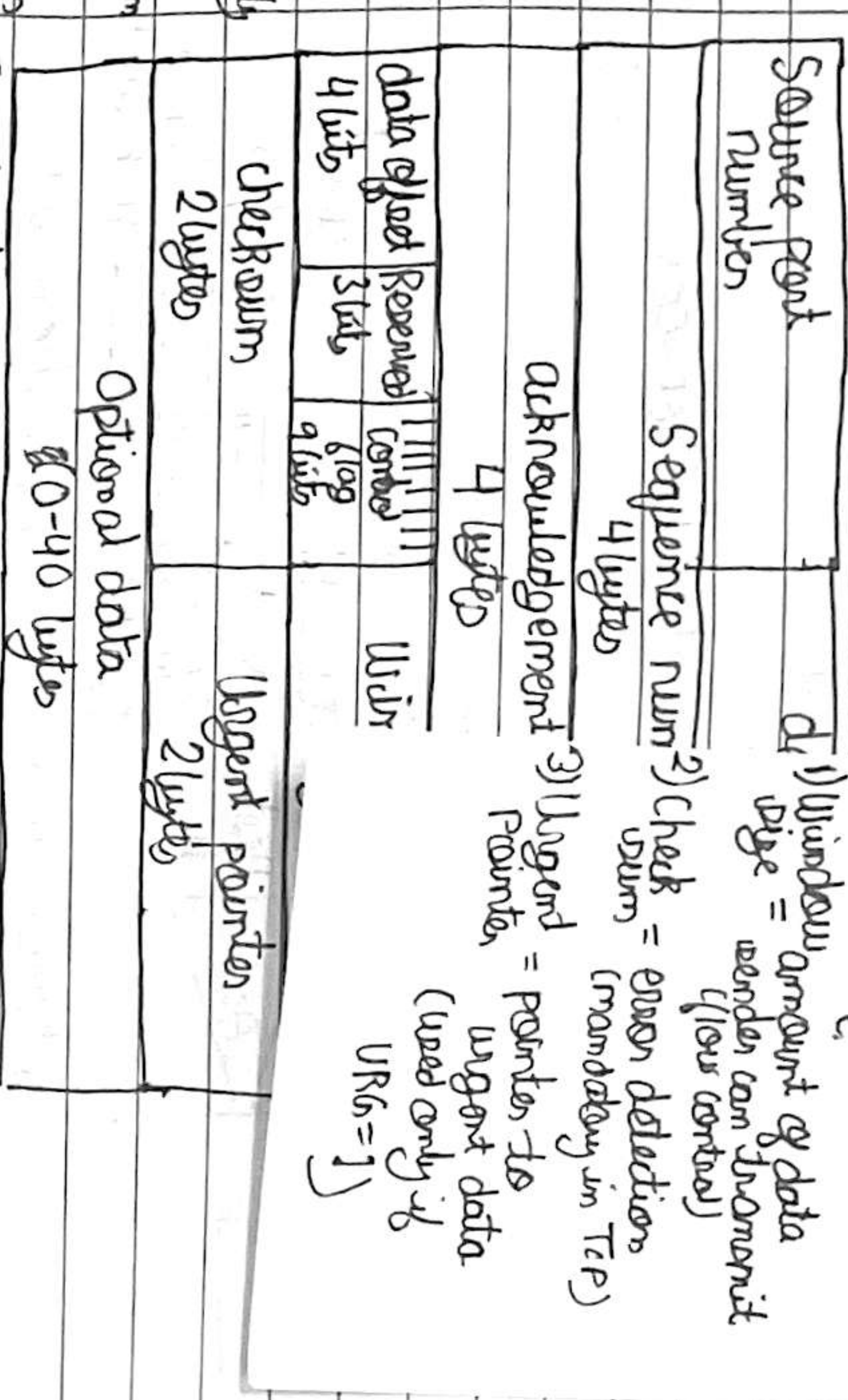
Both sides exchange ISNs (32 bit)

- Connection Termination (4 Step FIN Process)

- (i) Host A sends FIN → enters FIN-WAIT-1 A enters
- (ii) Host B sends ACK → enters CLOSE WAIT, enters FIN-WAIT-2
- (iii) Host B sends FIN → enters LAST-ACK
- (iv) Host A sends ACK → enters TIME-WAIT, then CLOSED

- TCP Header :

Other Key Fields -



- TCP Header Fields :

(a) Addressing

- Source Port (16 bit) - sender application
- Destination Port (16 bit) - receiving application

(b) Numbers

- Sequence Number (32 bit) - byte no. of first in segment
- Acknowledgement Number (32 bit) - next byte expected
- Header Length (HLEN)

4 bit field: value represents no. of 4 byte words in header

27 OCT 2025

Stream Control Transmission Protocol (SCTP) -

- Connection oriented transport layer protocol used to transmit multiple streams of data between 2 endpoints.
- Hybrid of TCP (reliability, connection oriented) and UDP (message oriented transfer)

- Applications:

- (i) Telephony over the internet
- (ii) Signaling in mobile networks

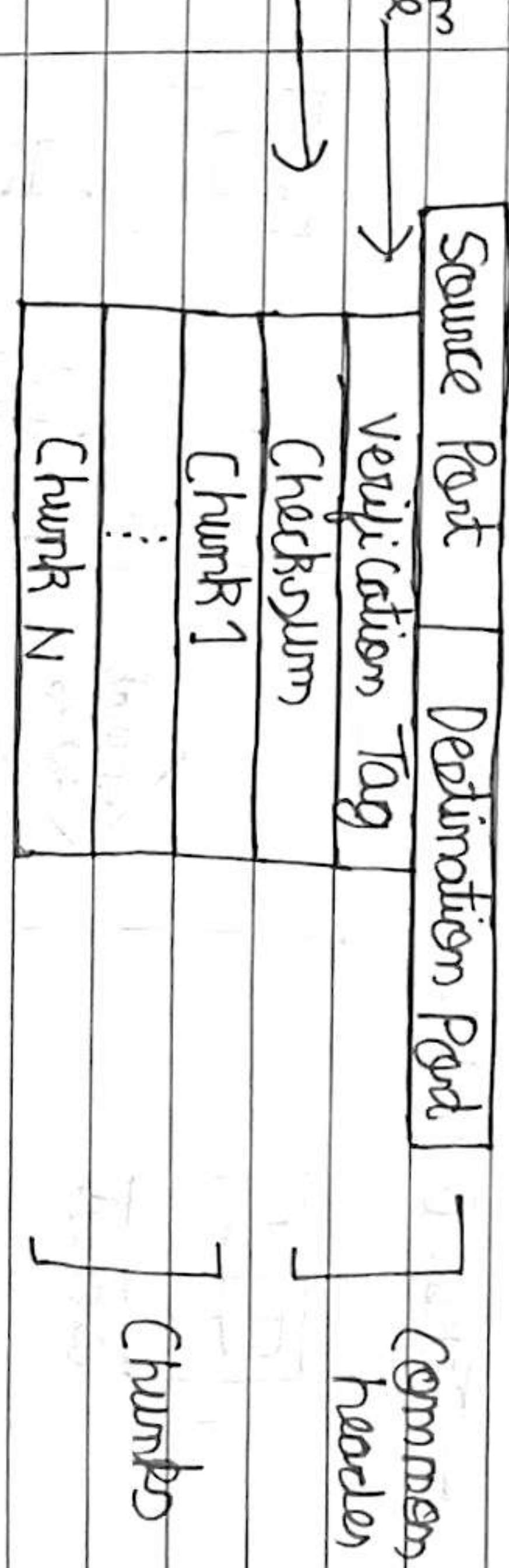
- Key Features :

- (i) Reliable & secure
- (ii) Message oriented
- (iii) Multihoming (allows end points to use multiple IP addresses for fault tolerance & path redundancy)
- (iv) Flow & congestion control (prevents overflow & network congestion)

Best Encryption algorithm?

- (v) Multi-streaming (supports multiple independent streams)
- (vi) Security (provides resistance against MITM DoS attacks)

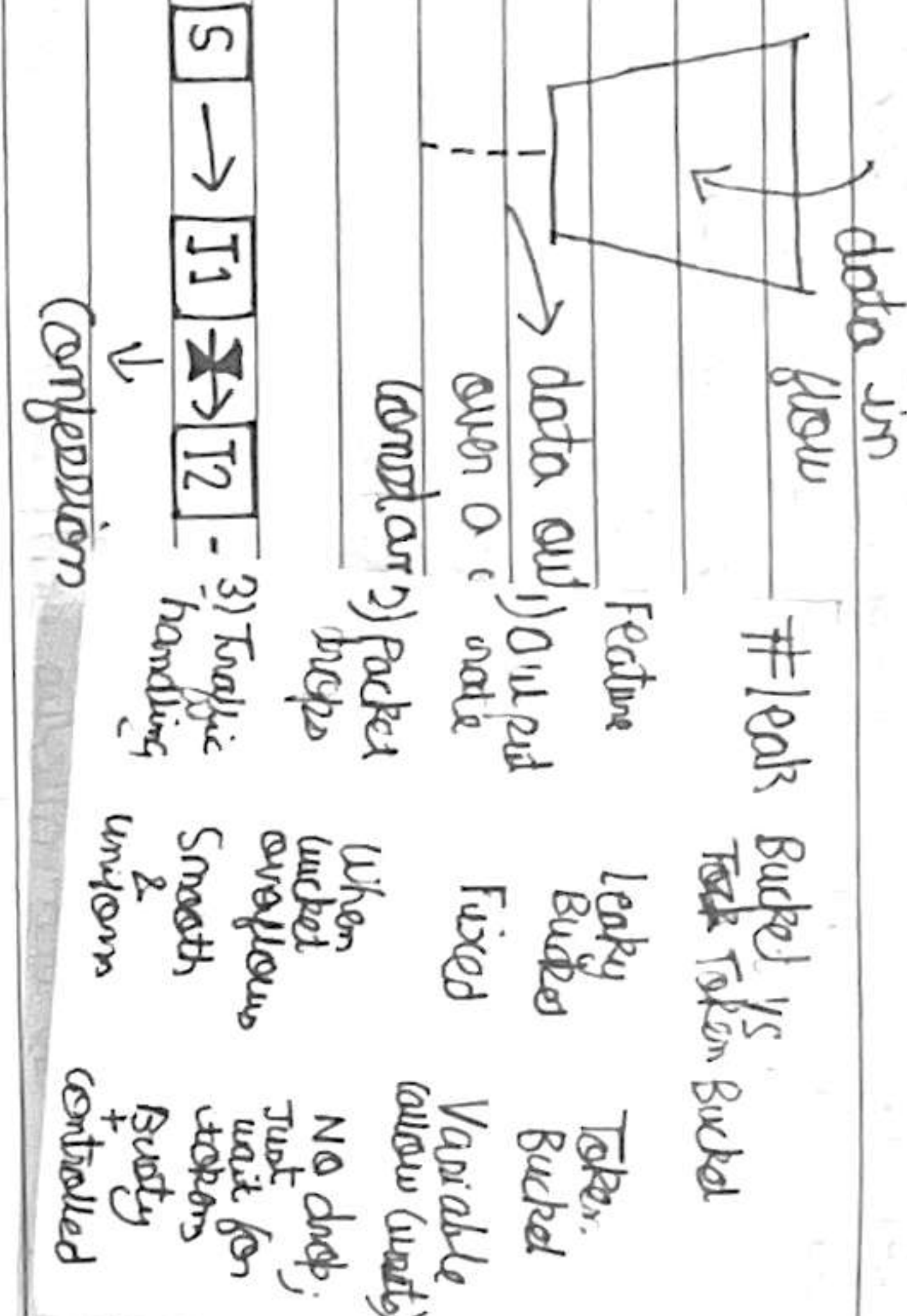
32 bit random value



- Common SCTP header is 12 bytes

Token Bucket = like medical appointment

leak Bucket algorithm -

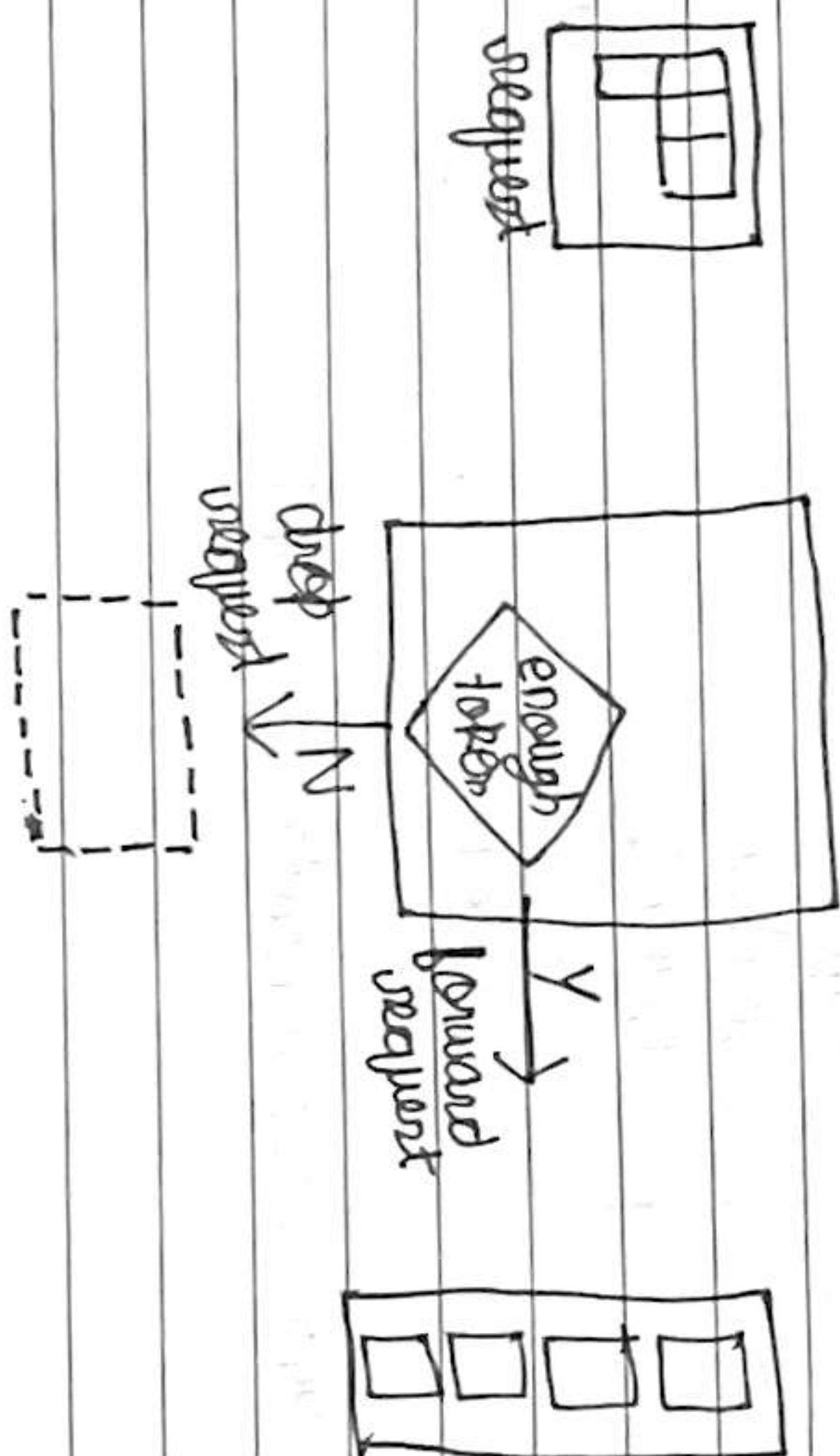


Quality of Service (QoS) algorithm -

- (i) Token Bucket
- (ii) Leaky Bucket

given $\rightarrow$ used to reduce the delay

Token Bucket -



TCP Delay Modeling -

Delay with analyzing the latency experienced by data sent over a TCP connection

Factors:

- (i) Transmission Delay
- (ii) Propagation Delay (time taken by last bit of the packet to reach the destination)
- (iii) Queuing Delay
- (iv) Processing Delay

$$T_{\text{total}} = T_{\text{propagation}} + T_{\text{transmission}} + T_{\text{queuing}} + T_{\text{processing}}$$

Principle of Reliable Data Transfer Protocol -

Assumes that the communication channel is ideal $\Rightarrow$ no packet loss, no corruption & no delays

- (i) No error detection and correction
- (ii) No acknowledgments
- (iii) No retransmissions
- (iv) In order delivery assumed (all data arrives in the exact order)
- (v) Flow control not required

Unreliable channel (bit errors & noisy channels)

Bit Error Channel:

- (i) Bit errors occur when noise flips bits during transmission
- (ii) TCP handles errors using checksums
- (iii) Retransmission is done when an error is detected
- (iv) ACKs confirm successful packet receipt
- (v) NAKs indicate corrupted reception

eg- TCP uses checksums

ARQ

Lossy Channel -

Packets may be completely lost during transmission due to issues like network congestion, buffer overflows or physical interference

- (i) Timeout & retransmission
- (ii) Sequence numbers
- (iii) Cumulative acknowledgments
- (iv) Sliding Window Protocol

Why Flow Control is needed?

- (i) Receiver buffer limitation
- (ii) Different processing speeds
- (iii) Network congestion

Types of Sliding Window Protocol?

- (i) Go Back N ARQ } UNIT-2
- (ii) Selective Repeat ARQ }

Timeout in TCP-

- (i) Retransmission timer (how long TCP should wait for ACK)
- (ii) Persistence timer (when TCP encounters a zero window condition)
- (iii) Keep alive timer (connection is kept alive when no data is being sent)
- (iv) Time wait timer (ensures one delayed packet does not interfere future connections)
- (v) Delayed acknowledgment timer
- (vi) Congestion control timer

Transactional TCP (T-TCP) -

- Designed to enhance performance in scenarios where speed & transaction are alive
- (i) Improved efficiency
- (ii) Explicit transaction boundaries
- (iii) Explicit transaction boundaries
- (iv) Optimized acknowledgments
- (v) Transaction Demarcation

Mobile TCP (M-TCP)

Designed to improve the performance & reliability in mobile environments

- (i) Connection continuity
- (ii) Smooth handling
- (iii) Change in IP address
- (iv) Reduced latency
- (v) Error recovery
- (vi) Support for multiple connections
- (vii) Retransmission

TCP Congestion Control -

1) Slow start:

Find available bandwidth with out overloading network

2) Congestion avoidance:

Prevent congestion by controlled increase

3) Congestion detection:

Network Security -

Goals:

- (i) Confidentiality
- (ii) Integrity
- (iii) Availability
- (iv) Authentication
- (v) Non repudiation
- (vi) DoS
- (vii) Man in the middle attack

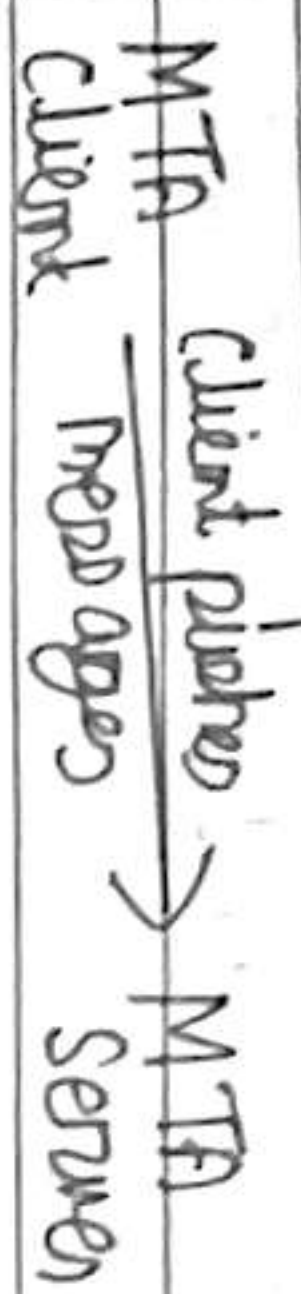
E-Mail -

Method of exchanging digital messages over the Internet using standardized communication protocols

Steps:

- (i) Sender uses a User Agent (UA)
- (ii) Email goes to Sender's Mail Server (MTA Client → MTA Server)
- (iii) Sender's MTA Server sends email across the internet
- (iv) Receiver's MTA Server stores the email
- (v) Receiver accesses email via MUA (if using LAN/WLAN)
- (vi) Receiver reads email through UA

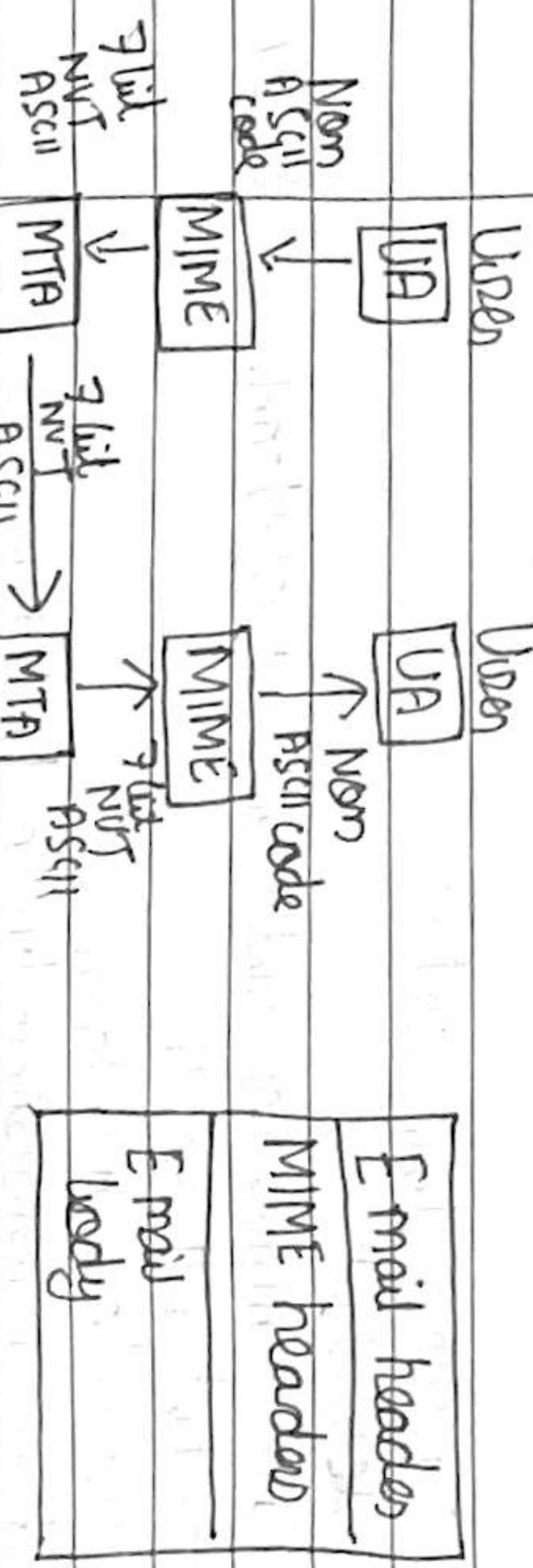
Push versus pull in email-



Services of user agent-

- (i) Composing messages
- (ii) Reading messages
- (iii) Replying to messages
- (iv) Forwarding messages
- (v) Handling mail boxes

MIME (Multipurpose Internet Mail Extension) -



World Wide Web-

Distributed client server service in which client using

a browser can access a service using a server

URL design 4 things:

- (i) Protocol = client server program that we need to access the web page. eg: HTTP & FTP

(ii) Host = IP address of the server

(iii) Port = 16 bit integer, pre-defined for client server app.

(iv) Path = identifies location & name of the file in underlying OS

HTTP-

(a) Non Persistent connections:

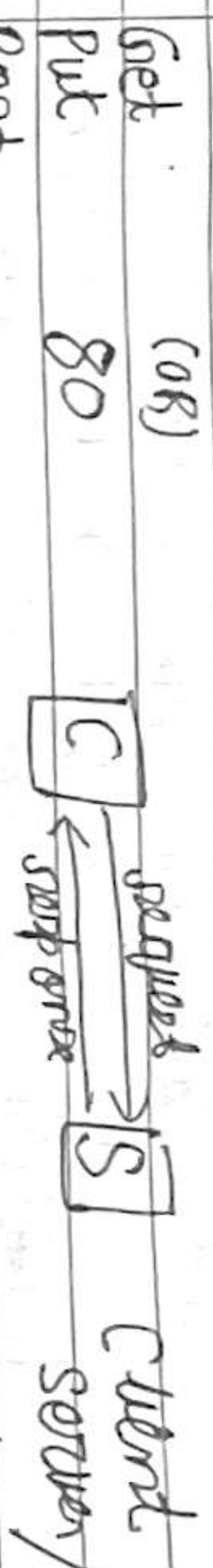
One TCP connection is made for each request/response

(b) Persistent connections:

Server keeps the server open for more requests after sending a response. Server can close the connection at the request of a client on timeout. Time & resources are saved here.

7 NOV 2025

HTTP => 8080



Master-Slave

Bluetooth-

- Wireless
- FHS
- 1Mbps or 3Mbps
- PAN

Firewall-

- Filters out networks/intenders
- Min 3 layers
- Work:

Accept
Reject
Drop

Domain Name System (DNS)

- Phonebook of the Internet
- Translates domain name to IP addresses
- Helps Internet users discover websites using human readable addresses.
- Runs over UDP & uses port 53
- Uses a hierarchical tree structure: 128 levels: Level 0 (root)
- Labels = Components: to level 127
- (i) Label: Each node in DNS tree has a label
- A string (upto 63 characters) that identifies it;
- next label = empty

(ii) Domain Name: Sequence of labels separated by dots that uniquely identifies a node

(iii) Domain: subtree within the DNS namespace

(iv) Distribution of: Process of storing DNS info across multiple servers to increase efficiency

(v) Zone: Part of DNS namespace for which a server is responsible, storing all records info in a zone file

(vi) Root server: Top level DNS server that directs queries to appropriate domain servers but does not store domain data itself

- ~~Domain~~ DNS in Internet:

(i) Generic domains = based on general purpose or type (eg. com, edu, gov)

(ii) Country domains = based on country code (eg. in, uk)

(iii) Inverse = used to map IP addresses to domain name, performing reverse of a normal DNS lookup

Directory Services -

collection of software, hardware, processes, policies and administrative procedures.

Directory services require:

(i) Directory information

(ii) Software servers storing this info

(iii) Software clients accessing it for use

(iv) Hardware running these clients/servers

(v) Access control policies

(vi) Software for maintenance & monitoring

- Earlier direct only services (X.400 email application)

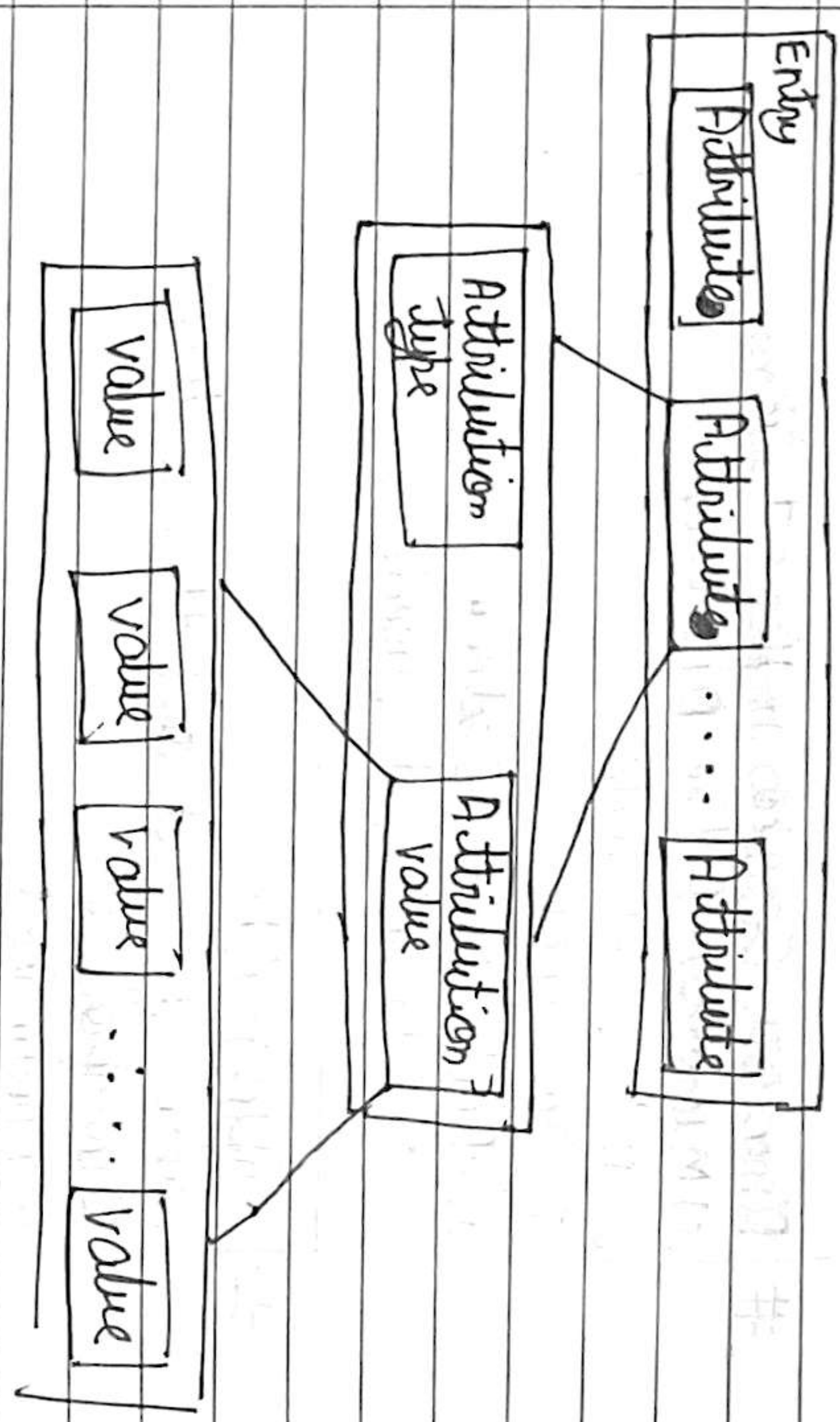
- Novus (X.500 introduced in 1988, now in 5th version)

- Implemented as a distributed database
- all network entities are implemented as objects with attributes

- Schema defines the directory (blueprint)

- X.509 became a common standard for authentication

Directory entry with attributes:



LDAP -

- lightweight alternative to X.500, designed to run over TCP/IP instead of heavier OSI stack

- Still uses X.500-based naming & structure

- LDAP operations:

- Authentication: open, bind, unbind

- Interrogation: search, compare

- Update: add, modify, delete

SSL = Simple Authentication and Security Layer

Quick Work
Page No.
Date
M T W T F S S

- LDAP security model ensures:

- Authentication
- Integrity
- Confidentiality
- Authorization

- LDAP authentication methods

- No authentication
- Basic authentication
- SASL, which supports stronger mechanisms like SSL/TLS

Directory Services implementations:

- Microsoft Active Directory
- NetIQ eDirectory
- OpenLDAP
- Apple Open Directory
- Novell Internet Directory

Cryptography:

- Science of secret or hidden writing

- 2 components:

- Encryption
- Authentication/Integrity

Cipher:

Method for encrypting messages

Plain $\rightarrow$ Encryption $\rightarrow$ Cipher $\rightarrow$ Decryption $\rightarrow$ Plain
text algorithm text algorithm text

Key A $\uparrow$ Key B $\uparrow$

Key = a string of numbers or strings
If Key A = Key B $\Rightarrow$ symmetric
If Key A $\neq$ Key B $\Rightarrow$ asymmetric

Symmetric Algorithms

- eg - Caesar Cipher

- Types -

(a) Block cipher

- encrypts data one block at a time
- used for a single message

(b) Stream cipher

- encrypts data one bit or more byte at a time
- used if data is a constant stream of info

- Key Strength:

- Size of the key
- Key length in bits (usually 48-448 bits)
- Set of possible keys for a cipher is called key space
- To crack the key $\Rightarrow$ brute force

*

- Caesar Cipher:

Each letter rotated by 3 letters

A $\Rightarrow$ D Z $\Rightarrow$ C M $\Rightarrow$ P

B $\Rightarrow$ E Y $\Rightarrow$ B N $\Rightarrow$ Q

- Substitution cipher:

(monoalphabetic)

Substitution cipher:

(a) Monoalphabetic cipher:

- Any letter can be substituted for any other letter

- Each letter has to have a unique substitute

A $\Rightarrow$ M (as we can't assign M to any other alphabet)

- Brute force approach would be too time consuming
- Statistical analysis would make it possible to crack the key

Knowing % of occurrence of different letters in the language commonly occurring 2-3 letter combination } ENIGMA

Quick Work
Page No.
Date
M T W T F S S

(b) Polyalphabetic Caesar Cipher -

(Vigenere Cipher)

- Uses a sequence of monoalphabetic ciphers in tandem

Eg. C_1, C_2, C_3, C_4, C_5

Plain text A B C D E F G H I J K L M N ...

$C_1 (K=6)$ F G H I J K L M N O P Q R S ...

$C_2 (K=20)$ T U V W X Y Z A B C D E F G ...

Bob, I love you. Alice

Gnu, I etox dhz. Tenvy

(c) Using 0 key to shift alphabet.

- Using a key for the algorithm & shift the alphabets

- If key = WORD, we will shift all the letters by 4 & remove the letters w, o, r & d from the encryption

- We have to ensure that mapping is 1-1

- No single letter in plain text can map to 2 different letters

- No single letter in cipher text can map to 2 different letters

Plain text A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
 C_1 W O R D A B C E F G H I J K L M N P Q S T U V X Y Z

Diagram showing mapping from Plain text to C_1 with arrows indicating shifts.

Transposition Cipher -

Columnar Transposition

- Rearrangement of characters on the plain text into columns

THIS I	TS SO H
SAME S	O A N I C
SAGE T	H A P A S C
OSHOW	I R S T C
HOWAC	I M G H C
OLUMN	
ARTRA	
NSPOS	
ITION	
WORK S	

Plain text

Basic Transpose

Write the column into new wire (left to right) while maintaining the dimensions (10 rows, 5 columns)

Shannon's characteristics of Good Ciphers -

- 1) Required secrecy should match the effort needed for encryption/decryption
- 2) Key & algorithms should be simple
- 3) Implementation should be easy & efficient
- 4) Errors in encryption should not spread & corrupt more data
- 5) Ciphertext should be similar to plain text size (not much larger)

Properties of Trustworthy Systems -

- 1) It is based on sound mathematics
- 2) It has been analysed by competent experts & found to be sound
- 3) It has stood the 'test of time'

- Resistance to ~~sup~~ tamoxifen is directly proportional

- ## # Data Encryption Standard (DES) -

- ## # Encryption ~~Summary~~ Algorithm Summary

Quick Work

Page No. _____

Date: _____

- Any exposure to the secret key compromises

- ## # Asymmetric Enolisation -

- 四

Public key	Private key

- Typed:

(a) PSA :-

- Both public & private keys are interchangeable

Variable Key size (512, 1024, 2048 bits)

Most popular public key algorithms

(b) EJ Gamal:

Variable Key Size (512 or 1024 bits)

less common than RSA used in protocols like PGP

limitations of Polymeric encryption -

- Efficiency is lower than symmetric algorithms
- Potential for man in the middle attack
- Problematic to get the key pair generated for the encryption

→ A hacker can create a fake key pair, share the public key while pretending it belongs to someone else, intercept the encrypted messages, read them & then re-encrypt them with the real secret key => making others think they are secure

Session-Key Encryption -

- Used to improve efficiency
- Symmetric Key is used for encrypting data
- Asymmetric Key is used for encrypting the symmetric key

Asymmetric encryption protocols:

- 1) Pretty Good Privacy (PGP)
- 2) Secure/Multipurpose Internet Mail Extension (S/MIME)
- 3) Secure Socket Layer (SSL) and Transport Layer Security (TLS)

Key Agreement -

- Method to create secret keys by exchanging only public keys
- eg.

- (i) Bob and Alice exchange their public keys
- (ii) Bob uses Alice's public key + his private key to create a session key
- (iii) Alice uses Bob's public key + her private key to create the same session key
- (iv) A key agreement algorithm ensures both derive the same shared key
- (v) No key needs to be sent between them

Diffie-Hellman Algorithm -

- Finds Key Agreement algorithm
- Inception of public key cryptography
- No exchange of secret keys
- Man in the middle attacks avoided

Authentication -

- Process of validating the identity of a user on the integrity of a piece of data
- 3 tech that provide authentication:

- Message digests
- Digital signatures
- Public Key Infrastructure

- 2 types of user authentication:

- Identity provided by a remote on application participating in a session
- Sender's identity is presented along with a message

Message Digests -

- Acts as a fingerprint for a document
- Ensure that data has not been altered
- Created using hashing
- Hash functions are one way
- Infeasible to create 2 different messages with the same digest
- Common hash algorithms: MD5 and SHA/SHA-1 (128 bits) (160 bits)

Message Authentication Codes -

- A message digest created with a key
- Creates security by requiring a secret key to be possessed by both parties in order to retrieve the message

Message $\longrightarrow$ Message Digest $\longrightarrow$ Digest

Algorithm $\uparrow$

Secret Key

Password Authentication -

- Secret character string only known to user and server
- Message digests (commonly used for password authentication)
- Stored hash of the password is a lesser risk

Authentication Protocols -

- Set of rules for secure communication between user and server
- Key techniques:
 - (a) Two-way shared password (hashed)
 - (b) Challenge response (like CHAP/CHN)
 - (c) Timestamp
 - (d) One time password (OTP)

Kerberos -

- Authentication service that uses symmetric key encryption & a key distribution center (KDC)
- Server stores symmetric keys of all users and maintains access ~~privileges~~ privileges for each user on the network

Personal Tokens -

- Hardware devices that generate unique values used with passwords for authentication
- Types of tokens:
 - Storage token
 - Synchronous OTP generation
 - Challenge response token
 - Digital signature token

Biometric Authentication -

- Uses certain biological characteristics for authentication
- eg.
 - Fingerprint recognition
 - Voice "
 - Handwritten "
 - Face "
 - Retinal ~~scan~~ / Iris Scan

Digital Signatures -

- A data item which accompanies on it is logically associated with a digitally encoded message
- 2 goals
 - (i) guarantee of the source of the data
 - (ii) Proof that data has not been tampered with

Digital Certificates -

- Signed ~~statement~~ statement verifying that a public key belongs to a specific party
- Signed by a trusted Certificate Authority (CA)
- Top level (root) Certificate must be self signed
- Affixance Verifies is a well known industry CA

Certificate Chaining -

- One certificate is signed by another whose public key is certified
- Works like passport signed by the govt proving identity
- Contains a person's public key + identity info, signed by a CA's private key
- CA's public key can verify (decrypt) the certificate
- Trusted signer is the CA